# Please add comments using ctrl+m -Chandeep 3/5/09 7:35 PM

this is from our Planet Action proposal which got us all the SPOT imagery worth about E300,000 from. PLEASE move items around or suggest to delete items which aren't in the usual proposal format-Chandeep 3/5/09 7:21 PM

*-=== Applicant Program Proposal
Chandeep Corea, Center for Conservation GIS of the Sri Lanka Wildlife Conservation Society.
chandeepc@slwcs.org or cc.suro@gmail.com

*-Title of your Project [PTITLE]?: Imagery for Land Use and Forest Cover GIS Analysis Project as support for Climate Change, Human Elephant Conflict, Wetland Conservation, Montane Tropical Forest Conservation, Community Development and Research Projects of the SLWCS Conservation GIS Center

*-PROJECT DOMAIN: put an "X" next to the relevant domain:
x__vegetation, biodiversity & ecosystems
x__human dimension & habitation
x__drought, desertification and water resources
x__oceans

*-EXECUTIVE SUMMARY of your proposal [PSUMMARY]?:
It is essential that we study the changes in land use around the world to enable us to make better appraisals of the trends in forests that influence their role in climate and the global carbon budget (http://www.whrc.org/programs/landcover.htm) Asian elephants have been recognized as a critically endangered species due to the vast deforestation occurring in Asia. They are further limited by the access to water and adequate forage. This brings them into constant conflict with the increasing rural population who are dependent on largely small scale agriculture and farming. In Sri Lanka alone, over 80% of forest areas have been decimated due to agriculture, irrigation, industrialization, urbanization, and logging. This destruction is due in many cases to the lack of effective landscape-level planning of necessary development activities. During the years 2004 to 2005 alone, the forests in Sri Lanka declined at a rate of about 1.5% per annum and cumulative forest loss since 1990 was 35%. As a result of habitats being frequently converted to agricultural lands, conflict between humans and elephants continues to increase at an alarming rate. The effect on climate caused by change of land use due to agriculture and human use along with the changes caused by high populations of elephants in limited areas needs to be studied[CC1] .

*-PURPOSE of your Imagery/GIS project [PDESGIS]?:
The imagery will be used by a Conservation GIS Center that we are establishing to integrate RS/GIS technology into both national and community-level planning to achieve measurable outputs on sustainable conservation of natural resources.

o Study the change of land use/forests in Sri Lanka through the recent past at a village level
o Use the above data to analyze the impact on climate change
o Create resource-use maps for planning of SLWCS Research activities
o Develop community-level maps for villages to better manage their own resources
o Publish fine-level GIS data on communities,  resources and human elephant conflicts in the project areas
o Apply GIS methods at the field level
o Create baseline GIS/RS for conservation planning to achieve the broader goals of the SLWCS projects.
o Make predictive models to look at elephant populations, habitat change and climate change at a village scale, to look for regional spatial relationships that could lead to solutions. For this high resolution imagery is needed annually and in each season.
o Conduct a comparative analysis of the  GIS-based elephant studies in Sri Lanka and Asia.

*-METHODOLOGY of your Imagery/GIS project [PDESMTH]?:
See below (PDESOUT) for methodology and results as well as partner organizations.

*-RESULTS: describe expected solutions, strategies & outcomes [PDESOUT]?:
Multiband images and high resolution orthocorrected images will be used to create current land use maps and modeling deforestation trends over time. Stereo imagery and vector contour datasets will assist in the creation of Digital Elevation Models and metapopulation/risk analysis modeling. These will be made into maps and predictive models so that communities and government officials can easily comprehend the negative trends and also see the positive impacts of model communities' appropriate land-use practices implemented with community participation.

SLWCS will work with the Sri Lanka Metrology Departments Centre for Climate Change Studies to study how the change of landuse obtained through analyzing the SPOT imagery has affected the climate. Daily data on rainfall in Sri Lanka will be collected from the Meteorological Department in the period where Spot Imagery is obtained. Data on other meteorological factors will also be collected from AGROMET stations. The following statistical methods will be used.

- Descriptive analyses consisting of the computation of average, minimum, maximum, coefficient of variation and quartiles for yearly and monthly data
- Time series techniques to identify any trends, seasonal effects or cycles for yearly and monthly data
- Probability analyses viz., incidence of a wet week, a wet week after a wet week (W|W) for weekly rainfall totals; 20 mm, 30 mm; 40 mm and 50 mm
- Regression analysis to identify the significance of the trend in yearly and monthly data

One use of the imagery will be in Smart Land-Use Analysis using the Land-Use Conflict Identification Strategy (the LUCIS Model). It uses ESRI's ArcGIS geoprocessing framework, particularly ModelBuilder, to analyze suitability and preference for urban, agricultural, conservation, and other major land-use categories. The model helps users to build land-use scenarios and determine potential future conflict among the categories. Essentially it will allow us to analyze the following:

- The various land use changes in the area, e.g. reforestation projects, Irrigation Projects, and the various Tsunami rebuilding projects and how they impact on the environment.
- Provide a dynamic tool to assist and advice local and national government authorities on "best" land-use strategies
- A tool to monitor and evaluate conservation and development projects as well as other land use practices in the area and their environmental impact over the long term.

SLWCS is working with the Geo-informatics Society of Sri Lanka to share it's knowledge base and organize GIS training for conservationists. We will be distributing our findings through the International Water Management Institute, IUCN Asian Elephant Specialist Group (of which we are the chair for the Human Elephant Conflict group), and the Sri Lanka Association for the Advancement of Science, along with many other Sri Lankan and international organizations and at international scientific forums. We have formed collaboration with All One River GIS Services of Denver, CO, USA to assist in advanced GIS analysis. We are working with the University of Sri Jayawardenepura of Sri Lanka and forming partnerships with other universities to work with students on practical GIS projects at our project sites.

The Nature Conservancy (TNC) has worked in GIS based Tsunami projects with the SLWCS since 2004 and the TNC states that forest projects designed to mitigate climate change should meet rigorous requirements by encouraging the world's governments to create a flexible framework that incorporates the actions of developing countries with meaningful incentives to encourage the preservation and restoration of forests. This is exactly the use that SLWCS envisages through the SPOT imagery donation. It is essential that the work is done at a national level as the issues of climate change are at minimum a national scale for an island the size of Sri Lanka (68,000$km^2$) where initial research shows that a lot of forest die-back is caused by pollution from South India. Imagery needs to be at 2.5m or better resolution as the solutions will be at a local/village level, while looking at the issues at a regional or national level.

*-SCHEDULE: Summarize your project schedule, key dates and
key deliverables for the next year [PDESPLAN]?:

December 2008 - Included in Planet Action 2009 Calendar
November 2008 - Recieved prestigious Equator Initiative Prize from UNDP
February 2008 – National and Regional Conservation GIS training in Sri Lanka with SCGIS founder Leslie Backus and the Geo-Informatics Society of Sri Lanka (GISSL).
February 2008 – National and Regional Conservation GIS training in Sri Lanka with SCGIS Advisory board member and co-founder Leslie Backus and the Geo-Informatics Society of Sri Lanka (GISSL).
March 2008 – Hiring of permanent GIS staff member to SLWCS, along with 3 current researchers who will be getting GIS training in Feb.
April 2008-Aug 2008 – Classification of imagery and field correction. Collection of current Landuse data from external sources (SDSL, UDA, IWMI, MoE, CEA, DWC and FD –see below for full titles)
Sept 2008-Dec 2008 – Analysis of classified imagery and creation of initial models. Development of further plans

*-SERVICE- Which local groups are you involved with and how will your project help them [PDESHELP]?:

- Geo-Informatics Society of Sri Lanka (GISSL):

This is the professional body for people working in spatial sciences in Sri Lanka

- International Water Management Institute (IWMI):

They are working to improve the management of land and water resources for food, livelihoods and nature. winner of 2007 ESRI Global award.

- Department of Wildlife Conservation (DWC) and Forest Department (FD).

These are the main state bodies responsible for conservation and environmental monitoring.

- Mahaweli Authority of Sri Lanka (MASL)

The authority is responsible for landscape development in SL for agriculture, irrigation and hydro-electric power. They have requested SLWCS to assist them in integrating conservation aspects to long term landscape level management.

- Survey Department of Sri Lanka (SDSL) and Urban Development Authority (UDA)

The main bodies collecting surveying and spatial information.

- University of Peradeniya and University of Sri Jaywardenepura

Working with departments to assist students in research and conduct field based conservation projects as part of curriculum

- Ministry of Education and schools around Sri Lanka.

We are teaching school students basic GIS and spatial science in our pioneering “Outdoor Environmental Education Program”.

- UNESCO and Govt. of Sri Lanka

We have been asked to be their consultants on 10 year conservation centered development plans and have assisted and will continue to assist them in declaration of UNESCO World Heritage sites.

- Sri Lanka Academy for the Advancement of Sciences and the Sri Lanka Environmental Professional Association.

  Essential in the Strategic Environmental Assessment (SEA) Process as various policy level program and mega projects are assessed as the raw data and published findings will form one of the layers needed for comprehensive SEA's.

- PROPOSED (no official collaboration exists at present)

  Ministry of Environment (MoE), Central Environmental Authority (CEA) along with it's Climate Change Division as well as Sri Lanka Metrology Departments Centre for Climate Change Studies and the Sri Lanka Tourism Earth Lung Initiative

  *The data we collect will be crucial in defining policies for a wide range of subjects which relate to*

*-STAFFING- Short description of the people who will use
the Imagery and GIS products you requested & what background,
skills & training they posess relevant to those products.
[PDESPAX]?:

- ✓ ▪ Chandeep Corea – 12 years experience in marine, biodiversity, sociology and forestry research as well as experience in sustainable development and project cycle. Initially self-thaught in GIS and 2007 SCGIS Scholar for advanced GIS Techniques under ESRI.
- ✓ ▪ 3+ Researchers (sociology, biodiversity and elephants) trained in GIS by GISSL/SCGIS Ravi should we add Mangala and Padma as well as Wilson?-Chandeep 3/5/09 8:02 PM
- ✓ ▪ 2 Permanent GIS staff –MSc/BSc in spatial sciences
- ✓ ▪ GISSL –consultants trained in advanced GIS and image analysis.
- ✓ ▪ Consultant GIS technicians from All in One River GIS of Colorado, USA, University of Barcelona, Spain and Ministry of Environment, Lithuania -laura roman 3/7/09 1:09 AM
- ✓ ▪ Tanya Keen –Head, Google Earth Conservation Grants
- ✓ ▪ Leslie Backus – Co-Founder SCGIS and wildlife biologist with a focus on predictive and spatially explicit model building for the management of endangered and threatened species for the purpose of sustainable conservation practices.

*-SCIENCE STAFF or SUPPORT - Who helps you with scientific
questions, methods and problems, what is their scientific
background, and what do they do? [ODESMNT]?:

SLWCS USA, Australia, EU and Sri Lanka based consultant researchers with experience in statistics, elephant ecology, forestry, animal husbandry and development.

Based on over 10 years of experience developing and implementing conservation programs in Sri Lanka, we have realized the critical need to use advanced GIS analysis to make our conservation programs more effective.  We realize how important it is to develop projects at the landscape-level integrating community participation and biodiversity research. Advanced GIS technology is essential to develop the research and management strategies for these projects. The use of advanced GIS with imagery from SPOT/SCGIS will help to develop solutions that will not only save lives by mitigating HEC, but will also contribute to improving the conservation of elephants and other wildlife and their habitats. It will save both human and elephants lives. We will be able to explore the connections between climate change, land-use, people and elephants at a local, regional and national scale.

The Sri Lanka Wildlife Conservation Society is a small organization committed to conservation through community development in Sri Lanka. It is a 501c3 tax-free organization in the USA and a fully registered Non-Governmental organization in Sri Lanka. See www.SLWCS.org for more details.

*-PROJECT AREAS:

Attached separately.

*-PROJECT BUDGET

(6 months duration).

Imagery: Received Spot Imagery for project from Planet Action as grant

Software: Received ESRI (Arc INFO) and Google Earth Pro as grants
Hardware: Purchased 3 laptops, 2 dual/quad core computers, GPS units including GPS PDA's with funding from the Canadian International Development Agency and Society for Conservation GIS
GIS Technician: Rs. 60,000 per month for 6 months
Field Trips (7 days at a time) = Rs. 50,000 each (includes accommodation, meals and transport) into 8 field trips =
SLWCS Administration/Financial oversight and utilities = Rs. 40,000 per month =
Hardware: A3 Printer/Scanner - Rs. 120,000
Consumables: Batteries, large format printing, stationary, internet - Rs. 100,000
REQUESTED from Ministry of Environment = Rs. 300,000.00

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[CC1]as per the CFP we need to add in info on how this will be used in policy and also Clean Dev. Mechansm (CDM) -Chandeep 3/5/09 7:59 PM

[CC2] If I understood correctly, the ministry of environment with its invitation is seeking for research proposals in climate change topic, and it is concerned about "low income groups", that is, human population vulnerability to meteorological phenomena and climate dynamics. As I see the SLWS is mainly concerned about conservation of wildlife and less involved in human dimension of the problem, so this might be a weak point at the time of the selection by the ministry.The way this weakness could be solved out is by highlighting that the study of land use changes is vital for understanding natural resources available in Sri Lanka, so without the studies of the natural habitats in SL , the resources available for human population remain unknown as well.

Besides I would also add in the text some overview that climate change dangers in SL, ( sea level rise, raise of temperatures, greater incidence of hazards like storms, floods, droughts, etc.) will increase the conflict between wildlife populations and human settlements, due to ecosystems changing. Land use mapping, measuring climate data, monitoring wildlife population and natural resources ( for example wetlands) is essential for a accurate prediction of the effects that climate change (especially when also combined with demographics, agricultural, economical and statistical data it is the key for climate change human vulnerability prediction)

I am aware I am being vague, and not really getting into the project points, I just wanted to let you know what I think it's missing in this text-project when I read the invitation of the Ministry of Environment.
-laura roman 3/8/09 4:31 AM